DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

Nikhat

10/10

Course	E-Waste & Environmental Management	**Class**	TE INFT
Duration	45–60 minutes	**Marks**	10

Group Details

Group No.:	6	**Device Investigated:**	Wired earphones
Date:	2/9/2026	**Condition:**	Non-functional
Member 1:	Devangini Jadhav	**Member 2:**	Swapnil Bhabal
Member 3:	Dhanraj Mogera	**Member 4:**	Nikhat Momin
Member 5:			

Condition: ☐ Functional ☑ Non-functional ☐ Damaged ☐ Unknown

Activity Evidence

During the activity, photograph all group members actively participating in the Device Anatomy investigation. Print the photograph and attach it below. A posed group selfie is not considered activity evidence.

ATTACH PRINTED GROUP PHOTOGRAPH HERE

Photo caption: Our group investigating the anatomy and end-of-life characteristics of a Wired Earphone.

Group members visible: all

ACTIVITY CHALLENGE

You are an e-waste engineer. Your task is to investigate the material anatomy of an electronic product, identify recovery opportunities and environmental risks, and propose engineering changes that improve its end-of-life management.

Core question: Can you make this device more circular?

Part A — Device Profile

Device name	Vivo wired earphones
Primary function	Audio listening & Voice communication.
Why does this device become e-waste?	☐ Obsolescence ☑ Damage ☐ Upgrade ☐ Repair not economical ☑ Failure ☐ Other: Not Working

Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

#	Component	Function	Material(s)	Why is this material suitable?
1	Earphone cable/conductor	Carries electrical audio	Copper wires + enamel insulation	Copper is electrical conductive & flexible making it suitable for carrying signals.
2	Cable outer insulation	Protects internal wires & short circuits	PVC/TPE plastic	It is flexible, lightweight, electrically insulating & provides protection against bending & abrasion.
3	Inline PCB	connects & controls the microphone	FR-4/fiberglass board + copper tracks	FR-4 provides mechanical support & electrical insulation, while copper provides conductive paths for current.
4	Microphone	Converts the user's voice into electrical signal	Metal + Plastic + electret	The small microphone needs lightweight materials and conductive paths to be fitting inside the small controller.
5	Push-button switch	Controls function such as Play/Pause and answering call	Metal contacts + Plastic body	Metal provides reliable electrical contact, while plastic provides insulation
6	Speaker / earphone driver	Converts electrical audio signals into sound	Copper voice coil + permanent magnet + diaphragm.	Copper provides good conductivity for the voice coil; the strong permanent magnet produces the magnetic field.

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

Part C — Material Detective

Tick the major material categories present in your device:

☐ Ferrous metals ☐ Aluminium ☑ Copper ☐ Precious metals

☑ Engineering plastics ☐ Glass ☐ Ceramics ☑ Semiconductor materials

☑ Rubber/elastomers ☐ Other: Neodymium/permanent magnet, FR4/fiberglass, enamel.

	Material 1	Material 2
Name	Copper	Engineering plastic PVC/TPE
Where is it used?	Earphone cable conductors, PCB tracks & speaker voice coil.	Cable insulation, PCB/earphone housing and switch components
Why is it used?	copper has excellent electrical conductivity & is ductile/flexible	It is lightweight, electrically insulating & flexible & can be moulded into small shapes.

	Material 1	Material 2
Can it be recovered/recycled? How?		

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

Material	Component containing it	Potential value	Recovery difficulty
1. Copper	Earphone wires, PCB tracks	Low / Medium / High	Low / Medium / High
2. Neodymium	Speaker driver	Low / Medium / High	Low / Medium / High
3. Engineering plastic	Cable insulation, earphone housing	Low / Medium / High	Low / Medium / High

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No, the most valuable material is not always the easiest to recover because it may be difficult to separate from other components.

Part E — Hazard Detective

Identify components that may require controlled handling.

Component	Possible hazard	Risk if improperly disposed	Handling requirement
PCB	Toxic material	Soil / water pollution	E-waste recycling
Plastic cable	Harmful fumes when burned	Air pollution	Do not burn
Magnet	Small / Strong magnet	Difficult to dispose	Recycle properly
Copper wires	Sharp strands	Minor cuts	Handle carefully

Most environmentally concerning component: PCB

Why? It can contain harmful electronic materials that may pollute the environment

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

Component	Reuse	Repair	Refurbish	Recycle	Controlled Disposal
Earphone cable	☐	☑	☐	☑	☐
Speaker driver	☑	☐	☑	☑	☐
Inline PCB	☐	☑	☑	☑	☐
Plastic housing	☑	☐	☑	☑	☐

Justify ONE of your decisions:

Part G — Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

#	Existing design problem
Problem 1	The earphone housing is difficult to open, because the small parts are glued together.
Problem 2	Wires are permanently soldered to the PCB & speaker, making repair difficult.
Problem 3	Different materials such as plastic, copper, metal & electronic components are combined in a very small assembly.

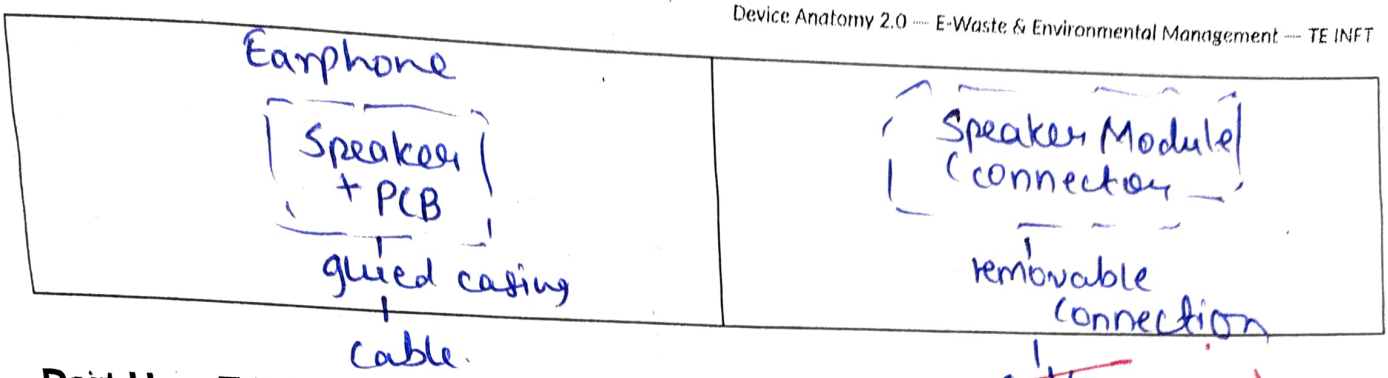
Now redesign it!

Propose three engineering modifications that would make the product more circular.

Existing Design Problem	Proposed Design Change	Sustainability Benefit
Difficult to open	Use screws/ snap-fit casing	Easy repair
Soldered wires	Use detachable connectors	Easy replacement
Mixed materials	Use separate modules	Easy recycling

Sketch your redesigned device/component — show at least one change that improves repair, disassembly or material recovery.

ORIGINAL	REDESIGNED



Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: Copper wires

DEVICE	CONSUMER DISCARD	COLLECTION	DISMANTLING	MATERIAL RECOVERY	NEW PRODUCT / RESOURCE

What could go wrong at any stage?

Improper disposal can cause pollution & loss of recyclable copper.

Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because...

It contains reusable and recyclable materials like copper, plastic & metals.

Which circular strategy would have the greatest impact?

☑ Repair ☐ Reuse ☐ Refurbishment ☐ Remanufacturing

☐ Recycling ☐ Material recovery

Why?

Repairing the earphones extends their life & reduces e-waste.

Part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? → Cu
2. Which material has the greatest recovery potential? → Cu
3. Which component is most difficult to recycle and why? → PCB
4. What is the single most effective design modification you propose? →
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

Final Group Verdict

The most valuable component is:	Copper wires
The component with the greatest environmental concern is:	PCB
The biggest design-for-recycling problem is:	Glued / closed casing & soldered connections
Our most important redesign recommendation is:	Detachable connectors & easy open casing

ONE-SENTENCE ENGINEERING VERDICT

We recommend designing for easy repair because it extends product life and reduces e-waste.

Assessment Rubric — 10 Marks

Criterion	Excellent (2)	Good (1.5)	Developing (1)	Marks
1. Engineering Anatomy	Accurate component + function + material analysis	Mostly accurate	Basic identification	/2
2. Material & Recovery Analysis	Strong material/recovery reasoning	Adequate reasoning	Limited reasoning	/2
3. Hazard & E-waste Assessment	Identifies risks with clear justification	Identifies major risks	Basic identification	/2
4. Design for Disassembly	Innovative, feasible engineering solutions	Relevant solutions	Basic suggestions	/2
5. Engineering Defence/Viva	Clear, evidence-based justification	Reasonable explanation	Limited justification	/2
TOTAL				**10**

Reflection

What surprised you most about the materials hidden inside your device?

The earphones contain several valuable materials in a very small device.

After this activity, would you still call this device "waste"? Why?

No, because many of its components and materials can be repaired, reused or recycled.

KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.